CS 619 Project Proposal

Forager: A Personalized Agent for Tracking AI News, Papers and Open-Source Tools

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1. Motivation

Users who want to stay at the frontier of several fields, such as an AI student who follows AI agents, video editing and image generation, must watch news sites, new papers from AI labs and trending open-source projects. Their problem is less finding information than organizing it: items repeat across sources, links across fields go unseen, and a short-term focus, such as a burst of interest in one model release, crowds out everything else. Feed algorithms learn users well but stay opaque, and AI assistants either demand long interest descriptions or learn silently. LLMs can understand semantic interests and synthesize many items, so they can help if users can see and steer them. Our goal is to keep users at the frontier of their chosen fields without losing sight of adjacent ones.

2. The App

Forager delivers a daily digest at a user-chosen time from three source types: tech news (RSS or a news API), papers (arXiv and AI-lab research blogs) and open-source releases (trending GitHub repositories and Hugging Face models). Each digest has two linked layers: a **quick brief** (5–8 key points by field, cross-field connections, and changes since last time, each citing its source items) and **full items** with summaries, links and a "why you see this" label. Users can also request syntheses such as "this week in image generation × video editing" and rate items.

3. Three Design Ideas

Idea 1 — Clarification-based elicitation (Week 1, intent expression). From one rough self-description, the agent splits interests into facets, asks follow-ups as clickable options, suggests adjacent fields, and shows an editable profile. Ad-hoc queries are clarified the same way (time range? timeline or comparison?). People recognize what they want more easily than they recall it. *Baseline:* free-text description.

Idea 2 — Consented, evidence-backed memory (Week 4, memory). Users opt in to each tracked signal (ratings, likes, clicks, reading time). The visible profile marks entries as stated or inferred, and as long-term or expiring short-term. The agent never updates silently; it proposes changes with evidence ("You rated three Sora items 5/5. Add video generation as a focus?") for the user to approve. *Baseline:* silent updates.

Idea 3 — Source specialists with a breadth-and-evidence critic (Week 6, multiple agents; also Week 2). A Manager delegates focused questions via `as_tool()` to three specialists with their own tools and instructions: a News, a Paper and an Open-Source Researcher (e.g., the Paper Researcher extracts a paper's key contribution; the Open-Source Researcher reports what a project does and how fast it is growing). A Critic, called by Python on every digest, checks breadth (no topic above 40%, every long-term interest within 7 days, a mix of source types, k justified exploration items) and evidence (each brief point cites items that support it), then returns PASS or ADJUST. Users see "interest match" vs. "breadth" labels, a weekly topic view and an exploration slider. Like crop rotation that keeps soil from monoculture, the Critic keeps the feed from narrowing: a relevance-driven recommender drifts narrow, and a separate agent with a different goal counters it with judgments a fixed rule cannot make. *Baselines:* single agent; rule-only topic cap.

Unlike the course's preference-memory example, Forager focuses on how preferences shape what users see: expiring short-term interests, breadth protection and traceable briefs.

4. Evaluation

We use the Week 6 harness on a fixed snapshot of news, papers and repositories. Code-judged cases check rules: topic share ≤ 40%, 7-day interest coverage, source-type mix, short-term weight decay, and valid citation IDs. LLM-judged cases, validated against human labels, check option-based clarification without invented preferences, evidence and approval before memory updates, use of consented signals only, and whether citations support each brief point. Idea 3 is compared with both baselines on topic diversity and relevance. A think-aloud study with 2–3 participants covers profile setup, reviewing an update proposal, and tracing and fixing a recommendation; Idea 1 is also compared with free text on setup time and self-rated profile accuracy.